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# Brief problem-solving therapy (PST) for women experiencing antenatal depressive symptoms: a randomised, controlled feasibility trial in an antenatal care setting in rural Ethiopia

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## Abstract

**Background** We adapted non-specialist healthcare worker (HCW) delivered problem-solving therapy (PST) to support improved treatment of antenatal depression in maternal healthcare platforms in rural Ethiopia. In this study, we aimed to evaluate the feasibility, fidelity, and acceptability of PST for antenatal depression and the procedures required for the implementation of a fully-powered randomised controlled trial (RCT).

**Methods** Participants were consecutive antenatal care (ANC) attendees in two primary healthcare (PHC)-based healthcare facilities, based on pre-established eligibility criteria. We employed a randomised, controlled feasibility trial design with two parallel groups. Participants in the intervention arm received four sessions of a contextually adapted version of PST; the control group received enhanced usual care (EUC) that involved routine antenatal care and information about sources of support. Assessments were conducted at baseline ($T_0$) and at 9 weeks after randomisation ($T_1$). Assessment at $T_1$ mainly comprised of preliminary clinical outcomes and hypothesized mediators. At $T_2$, feasibility parameters were assessed in addition to $T_1$ assessments. A recruitment rate of 80% of the expected sample size within a month, retention rate of ≥ 50%, provider competence of ≥ 60%, less than 10% adverse events and mean session duration of 30 min were assumed for feasibility success. We summarised participant characteristics, feasibility outcomes, and preliminary clinical outcomes using descriptive statistics.

**Results** After screening and the consent process, we randomised 50 eligible women to PST and EUC arms. Trial procedures such as participant recruitment and allocation concealment were feasible. Most (64%) completed all four sessions of the intervention. About 58% of HCWs attained expected scores on Enhancing Assessment of Common Therapeutic factors (ENACT) scale. No adverse events occurred during the trial. Supervision reports indicated that HCWs appreciated skills gained from PST and adhered to most of the PST content, except challenging women who reported not having problems or worries. However, they recommended pre-recorded videos of role play demonstrations and shorter trainer demonstrations, to increase trainees' opportunities to practise skills.

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**Conclusions** Participant recruitment, screening, randomisation, masking, as well as the intervention delivery process, were feasible. Future training should prioritise opportunities for HCWs to practise skills as much as possible.

https://pilotfeasibilitystudies.biomedcentral.com/articles/10.1186/s40814-021-00773-8

**Trial registration** The protocol was registered in the Pan-African clinical trials registry, (PACTR): registration number: PACTR202008712234907 on 18/08/2020; URL: https://pactr.samrc.ac.za/TrialDisplay.aspx?TrialID=9578. The protocol has also been published: https://pilotfeasibilitystudies.biomedcentral.com/articles/10.1186/s40814-021-00773-8.

**Keywords** Antenatal depression, Problem solving therapy, Feasibility trial, Ethiopia

## Key messages regarding feasibility

- The feasibility of PST was impacted by the following: (1) women attributing problems to situational factors, such as poverty or problems with their husbands, (2) women being distracted during sessions by accompanying children, (3) women's confidence to discuss their emotions in Amharic, while being audio recorded, (4) HCWs' focus on recording women's worries, and (5) the cultural practice of postnatal confinement limiting some participants from completing all four sessions or attending an outcome assessments.
- The feasibility of participant recruitment and randomisation was evidenced by our ability to approach 17 women per day, of whom nearly half met eligibility criteria. Randomisation to PST and EUC arms was feasible and acceptable, as were masking outcome assessors and investigators to allocation status, and baseline and outcome assessments 9 weeks later.
- Although this feasibility study was not powered to evaluate effectiveness, gross changes in mean clinical outcome scores such as GAD-7 and PCL-5 and hypothesised mediators (perceived social support, self-efficacy, mastery) favoured the PST arm. Each PST session lasted a mean 41 min: longer than intended. Techniques such as 'worry time' and relaxation exercises were appreciated, supporting the appropriateness of the PST model. Many (64%) women attended all intervention sessions. Participants rarely dropped out of the study, except those who moved away from the study site. Acceptability was optimised by integration of PST into existing ANC in PHC facilities. However, only 58% of HCWs attained expected ENACT (Enhancing Assessment of Common Therapeutic factors) scale scores, indicating lower than expected competency.
- A future efficacy trial should reconsider gestational age inclusion criteria in light of reduced post-natal attendance. Audio recording sessions for HCW supervision was not uniformly accepted in this setting. The content of PST sessions may require reduction to be feasible for widespread delivery in ANC.

## Background

Depression in the antenatal period contributes substantially to maternal morbidity worldwide [1]. In low- and middle-income countries (LMICs), depression is estimated to adversely affect about 16% of pregnant women [2, 3] and has been associated with functional impairment [4–6] such as reduced self-care [7–9], greater household food insecurity [10], and somatic complaints [5, 11, 12]. After childbirth, depression has also been found to be associated with delayed initiation of breastfeeding [13], poor mother-infant attachment, maternal and child undernutrition [13, 14], and poor child health [13]. In our previous work in rural Ethiopia, depression during pregnancy was associated with increased perinatal physical health complications [15], increased use of emergency healthcare [16], and unplanned (emergency) institutional delivery [17].

Although integrating detection and treatment of antenatal depression into maternal care platforms, particularly within primary healthcare (PHC) settings, has been recommended [5, 11, 12], it is not prioritised in many LMICs [18]. Evidence-based guidance recommends treating depressed women with antidepressants and/or psychotherapy in PHC-based maternal care services. However, antidepressants are more appropriate for moderate-severe depression [19, 20] and medication has low acceptability in pregnant women [21, 22]. Mild depressive symptoms not necessarily meeting criteria for major depressive disorder are associated with functional impairment and adverse effects on the woman and the foetus [23, 24] but can be treated effectively with psychological interventions [25–28].

Although Ethiopia is predominantly rural, mental healthcare remains highly centralised. Mental health specialists located in urban hospitals must prioritise treating people with severe mental illness, leaving a large treatment gap (96%) for perinatal depression [29]. The Federal Ministry of Health is seeking to expand access to mental health care in Ethiopia through integration in PHC and maternal care platforms [30] but is hampered by the lack of therapeutic options for antenatal depression. Culturally and contextually relevant and scalable psychological interventions are needed to address this gap.

The World Health Organization Mental Health Gap Action Programme intervention guide (mhGAP-IG) for PHC workers recommends several brief, manualised psychological therapies for people with depression, highlighting the need to adapt interventions to the cultural context [31]. In our formative qualitative work, pregnant and postpartum women and community HCWs perceived depressive symptoms as responses to situational challenges [32] which may thus be amenable to problem-orientated psychological interventions [33]. As previously detailed [32, 34], we selected the South African version of problem solving therapy (PST) [35], due to its suitability for empowering women in low-income, rural areas with problem solving skills and because it had been adapted for delivery by non-specialist health workers in PHC settings [36]. PST can be delivered by non-specialist, community and healthcare workers [27] in the context of routine maternal and child healthcare [25, 26]. In meta-analyses, PST has been found to be effective for treatment of depression in comparison to no treatment [37, 38]. Pragmatic trials of the South African version of PST among people with HIV and diabetes demonstrated significant reductions in severity of depressive symptoms at 12 months [39]. Furthermore, PST is simple, acceptable and effective in low literacy populations [37, 40–42]. Our adaptation process was guided by the Medical Research Council (MRC) framework for development and evaluation of complex interventions [43] and the 'ADAPT-ITT' approach [44] to adaptation. This adaptation process comprised theory of change workshops with stakeholders, adaptation workshops, in-depth interviews with women with a history of perinatal depression and HCWs [34]. Once adapted, PST was demonstrated to stakeholders through role play 'theatre testing', after which their feedback was incorporated into further adaptations [34].

MRC guidance recommends testing any adapted intervention's feasibility, acceptability, provider fidelity and preliminary effectiveness. Interventions should be acceptable to stakeholders with respect to culture, programme theory, timing, and human and material resource implications. Provider fidelity refers to the adherence of therapists to the planned material. In this study, we aimed to (1) assess acceptability, provider fidelity and feasibility of contextually-adapted brief PST for antenatal depression in rural Ethiopia, and (2) examine the feasibility of procedures of a randomised, controlled study design, to inform a future, fully-powered randomised controlled trial (RCT).

## Methods

### Setting

We conducted the study in purposively selected PHC facilities in Bui and Kella districts, located in the east Gurage zone of Central Southern Ethiopia, approximately 100–120 km south of Addis Ababa, the capital city. The official language of the region and in both districts is Amharic. In both study areas, the Programme for Improving Mental Health Care (PRIME) [45] study previously worked with local leaders and stakeholders to develop and implement a task-shared mental health care plan delivered in primary and maternal health care settings.

Participating PHC facilities were staffed by nurses, midwives, and health officers. Each facility was linked to about five health posts, community-based health facilities staffed by health extension workers (HEWs). HEWs are community-based HCWs who provide the first antenatal consultation, refer women for further antenatal care (ANC) at a health centre or primary hospital and maintain contact with women during pregnancy.

### Study design

We conducted a randomised, controlled feasibility trial with two parallel groups and two time point assessments: at baseline and nine weeks after randomisation [46]. We compared adapted PST with enhanced usual care (EUC) to examine the feasibility of procedures of a randomised, controlled study design, to inform a future, fully-powered RCT. A second feasibility trial, involving a sub-sample of trial participants who reported past-year exposure to intimate partner violence, was nested within the trial; as detailed in the published trial protocol [47, 48].

### Participants

Study participants were required to be pregnant women attending ANC services in PHC facilities. Women were eligible to take part if they (1) endorsed elevated and disabling depressive symptoms (scoring 5 or more on the locally validated Patient Health Questionnaire (PHQ-9) [49] and endorsed impaired functioning on the 10th PHQ-9 item about functional impact), (2) were aged 16 years and above, as this is the age at which married adolescent women become legally autonomous in Ethiopia, (3) were between 12 and 34 weeks' gestation of pregnancy, (4) spoke Amharic (the official regional language), (5) planned to live in the study area for at least 6 months following recruitment. The first eligibility criterion aimed to screen women for clinical symptoms on which the intervention aimed to work. Women below 12 weeks of gestation were not eligible because (1) pregnancy before 12 weeks of gestation is infrequently identified in this

rural setting because of cultural taboos surrounding early pregnancy disclosure and (2) it is not easy to distinguish early symptoms of pregnancy from depressive symptoms in this early stage of pregnancy.

Women were excluded from the study if they: (1) had a condition that impaired their capacity to understand the interview (e.g. were diagnosed with severe intellectual disability), (2) presented with acute medical illness or evidence of severe mental illness, (3) endorsed the ninth item of the PHQ-9 indicating risk of suicide and scored more than 17 on the suicide schedule of the Mini International Neuropsychiatric Interview (MINI) [50], or (4) required emergency treatment.

#### Participant recruitment and screening

We recruited consecutive women attending PHC-based ANC in two PHC facilities from late April 2021 for 20 working days. Research staff checked the initial eligibility of ANC attendees based on information available from their clinical records (gestation, age, co-morbid conditions, residence). Eligible women were invited to provide informed consent to an initial screen for depressive symptoms associated with disability using the PHQ-9 and the PHQ-9 tenth item on functional impact. This item asks 'Over the last two weeks, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?' Options are 'not difficult at all'=0, 'somewhat difficult'=1, 'very difficult'=2 or 'extremely difficult'=3. Women at risk of suicide (scoring over 17 on the MINI) [50] were excluded from the study and referred for review by a mental health-trained PHC worker [51] who employed the WHO mhGAP-IG [52] to assess the risk of suicide and provide an intervention. Research staffs were trained to facilitate women's attendance at services if referred; transportation costs to access care were covered by the study.

Any woman who was eligible to take part following the initial screen received written and verbal information about the study from research staff. Women either signed their consent or provided a witnessed thumb print if non-literate. Consenting participants underwent a structured baseline interview after receiving routine ANC. Participants' costs incurred to attend research interviews were reimbursed and time spent during research interviews was compensated. Time spent attending PST sessions was not compensated to avoid incentivising engagement with the intervention. HCWs were compensated for attending training, supervision, and delivering sessions.

#### Sample size

Fifty participants (25 in the intervention and 25 in the EUC group) were recruited for this feasibility trial. A sample size of 24–50 is recommended for feasibility studies [53]. The estimated sample size enabled us to detect a retention rate of 93% with a 95% confidence interval and a 5% margin of error [54].

#### Randomisation, allocation concealment, and masking

Women with disabling antenatal depressive symptoms were randomised to one of two arms (PST or EUC) using simple randomisation in a 1:1 ratio. The randomisation sequence was generated by an independent statistician, and group allocation was conducted centrally to ensure allocation concealment. When a new participant was allocated to PST, research staff informed a HCW trained to deliver PST. Intervention and control groups were anonymously coded so that data analysts remained masked to allocation status during analysis. Participating women and HCWs were not masked to allocation status due to the nature of the intervention; however, outcome assessors were masked to allocation status.

#### PST intervention

PST is a commonly used psychological intervention [38, 55] that has comparable efficacy to others such as behavioural activation and cognitive behavioural therapy, as well as to antidepressant medication [37, 56]. Assuming links between social adversities and depressive symptoms [57], PST teaches problem-solving and coping skills [57] in three phases. We adapted a South African model of PST in partnership with the original intervention developers (KS and BM) [35]. In the first phase, women were guided to identify the most important things in their lives. In the second phase, women identified a list of problems that challenge attainment of the previously identified priorities in their life before classifying problems into three categories: 'problems that are upsetting but not relevant to the most important things in my life' (Group A), 'problems that are important but insoluble' (Group B), and 'important and soluble problems' (Group C). In the last phase, women were encouraged to plan how to achieve the things they had identified as important in their lives. The details of the adaptation procedures and adaptations for this study have been published [34].

Our adapted PST model consisted of a PST providers' training manual, training slides, training guides, an in-session flip chart resource, and session record forms. Participants randomised to the intervention arm received four individual sessions from a trained, supervised HCW, along with EUC. Intervals between sessions ranged from a minimum of two days (for women whose delivery was considered imminent) to a maximum of two weeks, with four sessions being offered over a period of 8 weeks.

Trained HCWs delivered the intervention under the supervision of one Ethiopian clinical psychologist and one Ethiopian psychiatrist trained in psychological therapies. Five days of training in the adapted PST were delivered to HCWs. Training was spread over two blocks of weekdays: one block of two and one block of 3 days, to maximise staff ability to attend all days without disrupting their shift work. Care was taken to accommodate Ramadan, orthodox Christian Easter, and a general election. All clinicians and research staff followed the Ministry of Health directive on COVID-19, which comprised mask wearing, alcohol hand gel, and 2 m distancing, throughout the study, including when delivering PST sessions.

HCW training comprised two main components. First, a 5-day classroom course focused on counselling, communication, and PST skills, employing interactive methods like lectures, role plays, and small group activities. Second, participants engaged in four accelerated PST sessions with high-intensity supervision and feedback to improve their competency [58, 59]. In the accelerated delivery of the first set of sessions, consenting women were assigned to a recently trained HCW. Trainers supervised HCWs during accelerated delivery of these sessions, enabling them to discuss practical challenges [34]. Some HCWs reported that some women denied having any problems in their first session, preventing subsequent intervention steps from being followed. Trial standard operating procedures (SOP) required PHC staff to discuss the purpose and privacy of audio recordings to facilitate HCW supervision. After communication skill and manual adherence problems were identified during early cases, we organised psychiatrist-led refresher training on responding to women's emotions [34].

As part of monitoring the fidelity of intervention procedures, all intervention sessions were audio-recorded and a random selection was assessed for intervention fidelity and provider competency, by trained, independent female Ethiopian psychologists. A fidelity checklist was adapted to assess whether the intervention components were delivered. Independent psychologists also evaluated HCWs' competency using the Enhancing Assessment of Common Therapeutic factors (ENACT) scale [58, 59]. HCWs were expected to score 3 ('done well') out of the three levels: ('done well', 'done partially', and 'needs improvement') on each ENACT competency. Supervisors provided HCWs with feedback on audio recorded sessions focused on intervention fidelity, core PST competencies and communication skills [60].

### Enhanced usual care

Participants randomised to EUC received usual ANC (which focuses on advice about reproductive and family health) and information about sources of support. In addition, the Federal Ministry of Health in Ethiopia had prepared evidence-aligned guidelines on mental health for PHC and maternal care settings [61]. PHC staff are expected to detect mental health conditions and to provide basic mental healthcare (non-specific psychosocial care for all and supervised prescription of medication depending on severity) [62]. All HCWs participating in the trial had been trained in the WHO mhGAP-IG for a minimum of 5 days, as per the Federal Ministry of Health's scale-up plans. As the mhGAP-IG had not been implemented at scale in Ethiopia, this represented an enhancement in usual care [63, 64].

Participants allocated to EUC and their HCWs were informed about the results of screening and received a leaflet about well-being in pregnancy and sources of support. These women attended two contacts: the pre-randomisation baseline assessment and the follow-up assessment nine weeks post-randomisation.

### Baseline assessment

Baseline assessment measures are as follows: (1) demographic information and (2) preliminary clinical outcome measures and potential hypothesised mediating variables.

Demographic information comprised age, education, marital status, age at marriage, husband's occupation and educational level, and obstetric history, including parity, pregnancy intention, and gestational age, past psychiatric history, and the list of traumatic experiences scale. We used an item from the Ethiopian Demographic Health Survey which we had previously used in our setting [16] to assess pregnancy intention.

#### *Feasibility parameters*

In line with recommendations for the reporting of pilot studies [65], we identified several quantitative and qualitative indicators of feasibility and acceptability to guide decisions about what procedures to carry through to the full trial and which procedures or intervention content and delivery should be modified.

These included (1) feasibility of participant recruitment contextually defined as proportion of eligible participants invited to participate out of the expected sample size; (2) retention in the intervention (participants who completed all four sessions); (3) HCW competency (number of HCWs who demonstrated competent communication skills as assessed with the ENACT scale); (4) safety as defined by presence of serious adverse advents; (5) appropriateness of outcome measures and of the intervention, defined as number of missing items and mean session duration respectively; and (6) acceptability of the intervention and study procedures according to qualitative

data obtained from supervision notes. Quantitative data also included the standard deviation of outcome measures and time taken to complete baseline and outcome assessments. A recruitment rate of 80% of the expected sample size within a month, retention rate of ≥ 50%, provider competence of ≥ 60%, presence of serious adverse events of less than 10%, and mean session duration of 30 min were considered indicators of progression criteria to proceed to definitive trial.

We assessed the acceptability of intervention and trial procedures using qualitative data obtained from HCW supervision records and embedded supervisor notes in a session fidelity and completion checklist. Supervisors aimed to have in-person discussions with HCWs after completing HCW supervision records and independent fidelity checklists, based on session audio-recordings. Supervisors were the same two experienced mental health specialists: one female and one male Ethiopian psychiatrist and psychologist, who had delivered the 5-day PST training course.

We also conducted in-depth interviews with HCWs and a sample of women post-trial (to be reported in a forthcoming process evaluation). Qualitative data obtained from the supervision notes and session fidelity and completion checklists are reported in this paper.

#### *Preliminary trial outcomes and hypothesised mediators*

Assessments of preliminary clinical outcome variables and hypothesised mediators were conducted at two time points: pre-intervention (T0; baseline) and 9 weeks after randomisation (T1; follow-up). At protocol stage, we proposed a third assessment 3 months post-partum, but this was not carried out due to disruption by the coronavirus pandemic and civil conflict.

We piloted the potential primary outcome for a future, fully-powered RCT to assess feasibility of using the locally validated Amharic version of PHQ-9 [66]. PHQ-9 has been widely used as a clinical outcome measure of depressive symptoms [67, 68]. In Ethiopia, PHQ-9 has been validated for antenatal women [69] and in PHC settings in the neighbouring district of the current study. The optimal score cut-off indicating probable depression in this context has been identified as five or more in PHC [49]. At that cut-off, PHQ-9 had a sensitivity of 83.3% and specificity of 74.7% for detection of major depressive disorder. In previous studies, a 50% reduction in PHQ-9 depressive symptom scores [68] after 6–8 weeks was defined as treatment response.

We piloted potential secondary clinical outcome measures for a future RCT: Generalised Anxiety Disorder-7 scale (GAD-7 [70]), WHO Disability Assessment Scale (WHODAS [71]), post-traumatic stress disorder (PTSD) checklist for DSM-5 (PCL-5 [72, 73]).

We piloted the following measures of hypothesised mediators of PST's effects: a five item 'non-graphic language' scale for intimate partner violence (IPV) exposure [74] and the WHO multi-country IPV study screen [75], the WHO attitudes towards gender roles scale [76], a translated multicultural mastery scale [77], an adapted self-efficacy scale [77], the stressful life events scale [78] locally adapted Client Service Receipt Inventory (CSRI) [79] measuring the number of emergency healthcare visits and costs, and the Oslo Social Support Scale (OSS-3), previously used in Ethiopia [80].

#### *Data quality*

We trained research assistants prior to data collection. Baseline tools were completed on paper and subsequently entered into the electronic Open Data Kit (ODK) programme. Outcome assessments were directly entered into ODK at the time of data collection.

### Data analysis

#### *Preliminary clinical outcome measures*

We calculated descriptive statistics to summarise the preliminary clinical outcome assessments at both time points for PST and EUC arms using STATA version 16. For continuous outcomes, we calculated standard deviations to inform future sample size calculations for a fully powered RCT for all participants for whom data were available (complete case analysis). We described the proportion of each group meeting criteria for treatment success (a 50% reduction in PHQ-9 score at T1) for the two arms.

We summarised feasibility parameters such as acceptability, fidelity, and preliminary efficacy of the following: participant recruitment, randomisation, masking, assessment procedures, and intervention delivery using both numerical and text descriptions. We used descriptive tables and graphs to compare the two arms at baseline and follow up. Finally, we adhered to the CONSORT Checklist [65] to report our findings.

### Ethical considerations

Our feasibility trial protocol was registered on the Pan-African clinical trials registry, (PACTR): registration number: PACTR202008712234907 on 18/08/2020; https://pactr.samrc.ac.za/TrialDisplay.aspx?TrialID=9578. We received ethical approval from the scientific committee of CDT-Africa and the Institutional Review Board of Addis Ababa University, College of Health Sciences (IRB reference: 049/19/CDT). All participants provided written or independently witnessed thumbprint informed consent. Clinical records and consent forms were stored in a locked research office for confidentiality. Research staff monitored adverse events,

by telephone, by following up unresponsive participants who had dropped out.

## Results

### Feasibility of participant recruitment and randomisation

Feasibility of recruitment was evidenced by research assistants approaching a total of 335 potentially eligible pregnant women over 20 working days (17 women per day). Nearly half ($n=154$, 46%) of women approached met eligibility criteria after pre-screening. The remaining 181 women were ineligible (48%: 87 were in late gestation, 41 did not plan to continue living in the area, 37 did not speak Amharic, 23 were too unwell, 2 did not understand the information provided and 1 was under 16 years old (Fig. 1).

Of the 154 potentially eligible women who were screened, 56 women scored below five on the PHQ-9, 11 women were excluded because recruitment targets for this part of the trial had been met, and 12 declined to participate. Finally, $n=25$ consented women were randomised to the third arm in the nested study (PST adapted for women experiencing IPV). The remaining 50 consenting, eligible women were randomised to the two parallel arms. Among women randomised to PST arm, 64% attended all intervention sessions.

Almost all ($n=49$, 98%) participants were married, 38% had completed primary level education, 56% were homemakers, 40% were multiparous, 76% had intended to become pregnant, and 65% had experienced one or more stressful life events within the last six months (Table 1). The average household size was 4.1 and 4.5 persons for PST and EUC arms respectively. The average age at the time of marriage was 20.2 and 20.0 years for PST and EUC, respectively. The average age of participants was 27.1 and 26.2 years in the PST and EUC arms respectively.

### Feasibility of data collection procedures and preliminary clinical outcome measures

The layout, instructions and order of assessment tools were acceptable. There were no privacy or ethical barriers to screening participants or carrying out assessments. Allocation concealment was feasible and it was possible to mask outcome assessors and those carrying out data analysis. No data were missing from baseline and preliminary outcome assessment measures.

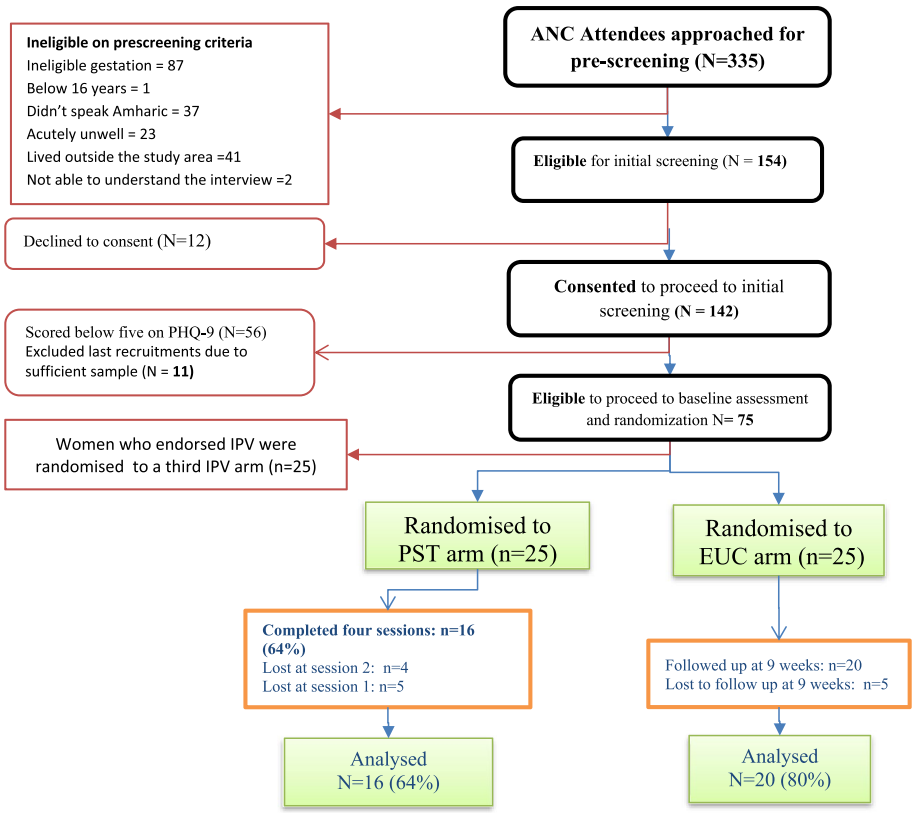



**Fig. 1** CONSORT participant flow diagram to show

**Table 1** Participant socio-demographic characteristics at baseline

| | | | Allocation to study arms | |
|---|---|---|---|---|
| Socio-demographic characteristics | | *N* = 50<br>Total<br>*N* (%) | *N* = 25<br>PST<br>*n* (%) | *N* = 25<br>EUC<br>*n* (%) |
| Marital status | Currently not in a relationship | 1 (2) | 1 (4) | 0 (0) |
| | Monogamous marriage | 48 (96) | 24 (96) | 24 (96) |
| | Polygamous marriage | 1 (2) | - | 1 (4) |
| Educational status | Non-literate | 12 (24) | 7 (28) | 5 (20) |
| | No formal education | 4 (8) | 0 (0) | 4 (16) |
| | Primary level only | 19 (38) | 14 (56) | 5 (20) |
| | Secondary level only | 8 (16) | 3 (12) | 5 (20) |
| | Higher education | 7 (14) | 1 (4) | 6 (24) |
| Occupation | Homemaker | 29 (60) | 16 (64) | 14 (56) |
| | Daily labourer | 3 (6) | 2 (8) | 1 (4) |
| | Farmer | 4 (8) | 2 (8) | 2 (8) |
| | Merchant/trader | 7 (14) | 5 (20) | 2 (8) |
| | Self/government employee | 5 (10) | 0 (0) | 5 (20 |
| Parity | Nulliparous | 18 (28) | 7 (44) | 11 (44) |
| | Primiparous | 12 (32) | 8 (16) | 4 (16) |
| | Multiparous | 20 (40) | 10 (40) | 10 (40) |
| Pregnancy intention | Unintended | 12 (24) | 6 (24) | 6 (24) |
| | Intended | 38 (76) | 19 (76) | 19 (76) |
| Stressful life events | None | 17 (34) | 7 (29) | 10 (40) |
| | One or more | 32 (65) | 17 (71) | 15 (60) |
| Summary of continuous variables | | | (Mean, SD) | (Mean, SD) |
| Age | | | (27.1, 6.2) | (26.4, 6.1) |
| Age at marriage | | | (20.2, 3.4) | (20.0, 3.5) |
| Household size | | | (4.1, 2.0) | (4.5, 2.5) |




**Fig. 2** Psychosocial outcome measures across study arms

Preliminary clinical outcome measures are presented in Fig. 2 for PST and EUC arms. Out of women followed up, a greater number of women in the PST arm (67%) fulfilled the definition of treatment response (50% reduction in PHQ-9 score) compared to women in the EUC arm (60%) (Table 2). However, more women in the EUC arm (40%) no longer met criteria for depression than women in the PST arm (33%) post-intervention. There was no consistent difference between arms in terms of response, remission, or mean PHQ-9 score. At follow up, mean

**Table 2** Change in preliminary outcomes across study arms

| Preliminary outcome | Assessment time | PST arm: mean, SD | EUC arm (mean, SD) |
|---|---|---|---|
| PHQ-9 score | Baseline ($n=25$) | 8.48, 2.22 | 8.16, 2.79 |
| | Outcome ($n=18$) | 4.60, 5.30 | 3.80, 5.12 |
| Generalised anxiety disorder-7 scale score | Baseline ($n=25$) | 9.72, 4.54 | 9.32, 3.01 |
| | Outcome ($n=18$) | 9.11, 2.47 | 9.70, 5.66 |
| WHO disability assessment scale score | Baseline ($n=25$) | 25.04 5.98 | 24.84, 5.94 |
| | Outcome ($n=18$) | 22.72, 9.63 | 23.35, 9.09 |
| PTSD symptom checklist score | Baseline ($n=25$) | 9.84, 15.23 | 5.36, 6.14 |
| | Outcome ($n=18$) | 4.00, 5.87 | 7.65, 12.08 |
| Non graphic language IPV scale score | Baseline ($n=25$) | 2.52, 3.11 | 2.68, 2.35 |
| | Outcome ($n=18$) | 2.50, 3.70 | 2.65, 4.28 |
| Perceived social support | Baseline ($n=25$) | 10.92, 1.93 | 10.60, 2.14 |
| | Outcome ($n=18$) | 11.17, 2.00 | 10.65, 1.89 |
| Efficacy of coping depression scale score | Baseline ($n=25$) | 13.64, 5.85 | 13.76, 5.79 |
| | Outcome ($n=18$) | 14.72, 5.94 | 16.4, 4.56 |
| Multicultural mastery scale score | Baseline ($n=25$) | 31.88, 6.72 | 33.28, 6.83 |
| | Outcome ($n=18$) | 35.05, 5.42 | 32.9, 8.38 |
| Attitude towards gender roles scale score | Baseline ($n=25$) | 27.08, 21.41 | 25.72, 6.48 |
| | Outcome ($n=18$) | 23.39, 4.30 | 25.20, 5.04 |

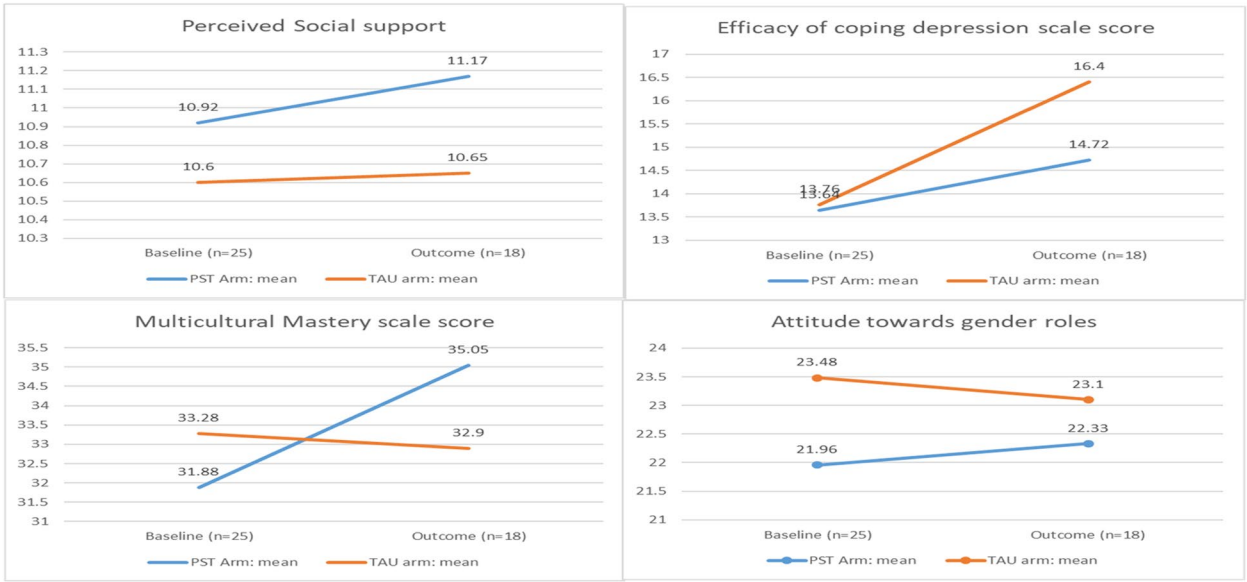



**Fig. 3** Hypothesised mediators across study arms

**Table 3** Time taken for each session

| Session | Session duration | | | |
|---|---|---|---|---|
| | Minimum | Maximum | Mean | Number of sessions timed |
| Session 1 | 0:21:05 | 0:56:00 | 0:40:45 | 10 |
| Session 2 | 0:37:00 | 1:09:00 | 0:47:48 | 7 |
| Session 3 | 0:16:14 | 0:58:42 | 0:40:53 | 15 |
| Session 4 | 0:34:00 | 0:45:00 | 0:37:39 | 4 |

PHQ-9 score was slightly higher among EUC participants than PST participants. Mean GAD-7 and PCL-5 scores were lower for the PST arm compared to the EUC arm at follow up.

The mean multicultural mastery score increased from baseline to follow up for the PST arm while it decreased for the EUC arm (Fig 3). Mean WHODAS scores decreased from baseline to follow up in both arms.

## Feasibility of intervention delivery

### *Appropriateness*

Supervision notes consistently indicated that coping strategies such as 'worry time', thinking stop and relaxation exercises were beneficial, supporting the appropriateness of PST from the perspectives of providers and service recipients. Nevertheless, the timing of childbirth prevented some participants from completing all four intervention sessions or attending outcome assessments nine weeks post-randomisation. This arose because of the cultural practice of postnatal confinement for 10–21 days.

All PST sessions lasted longer than the expected 30 min, especially session 2 (mean, 47.5 min) (Table 3).

### *Provider competency*

No staff withdrew from delivering PST at any stage. Training feedback indicated that HCWs generally appreciated the intervention and training methods. Audio-recorded session-based regular supportive supervision and feedback received from trainer/supervisors gave HCWs opportunities to discuss challenging cases. Only 58% of HCWs scored 3 on each ENACT item (Table 4). More than half of HCWs were independently rated as competent in verbal communication skills ($n=19$ sessions out of 36 rated); building trust ($n=30$ out of 35 sessions rated), empathy, warmth, and genuineness ($n=28$ out of 35 sessions rated) and promotion of realistic change ($n=33$ out of 35 sessions rated); and using local terms ($n=11$ sessions out of 14 rated). However, relatively few HCWs ($n=7$ out of 35 sessions rated) explained confidentiality to women, or asked for feedback and suggestions ($n=17$ out of 35). Scores were also low for exploration, interpretation and normalisation of women's feelings and assessing the effect of recent life events on women's well-being.

ANC staff were over-stretched during the COVID-19 pandemic at the time of training. For example, the trial coordinator reported her observation of two HCWs attending the training getting tired after night shifts. Supervisors noted that the quality of sessions and fidelity

**Table 4** ENACT scores across sessions (total $N = 36$)

| | Frequency for options of ENACT items | | | |
|---|---|---|---|---|
| Building relationship | **Needs improvement** | **Partially done** | **Well done** | **Total** |
| Verbal communication skills | 1 | 16 | 19 | **36** |
| Building trust | 0 | 5 | 30 | **35** |
| Explaining confidentiality | 20 | 8 | 7 | **35** |
| During intervention skills | | | | |
| Exploration, interpretation and normalization of feelings | 0 | 20 | 16 | **36** |
| Empathy, warmth and genuineness | 0 | 7 | 28 | **35** |
| Promotion of realistic hope for change | 0 | 2 | 33 | **35** |
| Use of local terms in delivering the intervention | 0 | 3 | 11 | **14** |
| Asking for feedback and suggestions | 2 | 16 | 17 | **35** |
| Assessment skills | | | | |
| Assessing recent stressful life events | 0 | 14 | 18 | **32** |
| Assessing problems solving skills | 2 | 6 | 26 | **34** |
| Harm to self, harm to others, and harm from others | 15 | 8 | 4 | **27** |
| Assessing potential community or social networks | 6 | 6 | 8 | **20** |
| **Total** | **46** | **111** | **217** | **374** |
| **Ratio of ENACT item options to total sessions** | **12.3** | **29.7** | **58.0** | **100.0** |

to the intervention increased as HCWs gained more experience.

HCWs reported that the use of face masks in a large room without microphones lowered the audibility of the trainers. Trainees also reported that trainer demonstrations of sessions could have been shortened, increasing time for trainee role plays and practical exercises.

Although many HCWs adhered to intervention procedures, supervisors reported that some HCWs needed practice to improve their basic counselling skills, including reflective listening, expressing empathy, and asking open ended questions without directly advising women:

> *"More practice of basic counselling skills; reflective listening, empathetic understanding has been advised. Directives/advising which doesn't exist in the PST protocol should be minimised."*
> *Supervisor*

Supervisors noted that some HCWs forgot to explain the content of each session, mixed up problems and the most important things in a woman's life or omitted to ask about these. One supervisor noted poor preparation:

> *"I have no concern about delivering PST but due to his low preparation when delivering this session, he has committed confusions when he explains issues. For example, he explains the 'most important things in your life as 'most significant concerns in life."*
> *Supervision notes.*
> *"She did not outline the three phases of PST at the beginning."*
> *Supervision notes.*
> *"[One] HCW was reading a lot from [the] flipchart which indicates limited preparation."*
> *Supervision notes.*

The supervisor noted that HCWs struggled to support women raising more intractable problems:

> *"I felt hopeless, unable to help her solve the problem she is in... I want to help/support those mothers even in the future. I mean what is the point of giving her the session if she is not going to have some sort of solution at the end of the day (she came for the second session looking nice though)."*
> *Supervision notes.*
> *"A mother who was unable to speak about her problem and come up with a solution, I was in a situation where I had to do all the talking. What to do in such cases?"*
> *Supervision notes.*

#### *Provider fidelity*

Trainer/supervisor notes indicated that several HCWs implemented PST as planned (*generally, depression during pregnancy was not mentioned, otherwise this HCW had done a great job to implement PST*). Supervisors praised HCWs' adherence to session introductions, such as introducing PST and discussing confidentiality. However, during independent fidelity assessment, assessors noted that some aspects of intervention content were not adhered to in certain circumstances, for example

when participants responded that they had no problems. Supervisors noted that this usually occurred when initial explanations and inquiries had been brief; they encouraged HCWs to explore women's lives further in such situations. One supervisor commented that some HCWs emphasised the six steps of problem solving and overlooked other therapeutic skills such as breathing exercises.

One HCW admitted that they skipped and had forgotten some elements and procedures of PST for the first woman they treated and queried the value of take-home activities for those who did not complete them. However, adherence improved for the second and third women they treated. Independent fidelity assessors noted that some HCWs struggled to categorise women's worries and concerns when they were all linked to socio-economic or interpersonal difficulties. Supervision notes indicated that thanking women for their attendance, helping them to identify their problems, defining problems and clarifying the purpose of audio recording were well done.

#### *Acceptability*

HCWs were compensated 2000 birr (37 USD) per woman who received the intervention, including training and supervision attendance. Pregnant women were reimbursed 70–140 birr for travel expenses incurred by attending sessions. Dropping out of the study for reasons other than moving away from the site was infrequent. Most HCWs were happy to take part in delivering PST and gaining skills:

> *"Because of PST, we admitted our deficiency of counselling skills and the training and practising [of] PST creates an insight for this. We are seeing progress-happy faces of woman, even with 2 or 3 sessions. So, we want to use [it] for our routine practice in the future."*
> *HCW views from supervision notes*
> *"Yes, she [HCW] found that the training is an additional useful skill to her routine activities. Since she is not familiar with this kind of psychosocial training, more rehearsal and preparation was recommended."*
> *Supervision notes*

#### *Engagement*

Supervisors observed that women sometimes became distracted by their children during sessions (*The babies of the women have been very distracting throughout the sessions*, Supervisor). Although Amharic is the official language of the region, some participants did not seem confident to speak in Amharic during sessions. One HCW worker mentioned that some women just cried, but most spoke up if spoken to in their local language:

> *"Generally, the session with this participant doesn't go very well which might be due to language barrier. She has been frequently talking in her local language and the HCW frequently requested her to speak in Amharic language. The HCW did not well internalise the structure of PST. Thus, he did not adhere to the structures of the PST as presented at the flip-chart."*
> *Supervision note.*

While several women spoke about their problems, some HCWs reported that women denied problems once the audio recording started: *I had to let go (didn't record the first session): one mother declined to take part, saying she has no problem*. HCWs pointed out that women's fear of their private matters being exposed to third parties was the reason for refusing to disclose their problems:

> *"They tell me lots of deep personal issues after we finish the session and the audio recorder is off. When we ask the problem they have most, people automatically go to the economical/financial problems they have. Maybe the more we talk to them and explain they will tell us the other problems we are looking for."*
> *HCW.*

Some HCWs asserted that some women required probing before opening up. Some HCWs reported confusion about the need to continue intervention sessions for women who engaged with problem solving in early sessions. Some participating women came to the health centre from remote areas and were in a hurry to complete sessions. For example, one HCW asserted that she worried about making the woman late and rushed during sessions while giving information and direction, against the manual content.

Fidelity checklists indicated that most HCWs encouraged women to join the discussion by thanking them. However, some HCWs did not provide encouragement. In the middle of the first session, HCWs engaged women in identifying, explaining and categorising their worries and concerns into three types of problems (*yes, she well engaged the woman here [in categorising the problem*]). Nevertheless, the supervisor commented that some HCWs were not actively listening and recording women's worries and problems:

> *"[The HCW] disregards some of the reported problems. At this point, she was directive and delivered guidance and advice instead of identifying worries and problems."*

The lowest levels of fidelity were for thanking the participant for attending sessions, choosing the problem that can be changed and working through the six problem solving steps, outlining the three phases of PST and explaining the take home activity. For example, one supervisor noted that some HCWs did not introduce and explain the six-step approach while others focused on lecturing about the six-step approach instead of practising it to solve women's identified problems.

#### *Safety*

No serious adverse events (SAE) or adverse events occurred during the trial.

## Discussion

The importance of tailoring existing evidence-based interventions for the context where they will be delivered is increasingly acknowledged [31, 43, 81]. We adapted PST for women with antenatal depressive symptoms in a rural Ethiopian setting [34] where there is a large mental health treatment gap linked to high social adversity. In this study, we demonstrated the feasibility of trial procedures and the possibility of integrating four sessions of adapted PST into ANC in rural Ethiopia. We explored contextual feasibility parameters, an essential step towards planning a future fully-powered RCT.

We recruited 100% of the expected sample size over 20 days and identified HCW competence of 58%, mean session duration of 41 min, and no serious adverse events. Key criteria for progression to a definitive RCT were fully met. The feasibility of participant recruitment and randomisation was also evidenced by identification of 17 potentially eligible women per day, of whom nearly half met inclusion criteria. Randomisation to PST and EUC arms as well as maintaining allocation concealment were feasible and acceptable, as were masking outcome assessors and investigators to allocation status. The faster rate of recruitment than expected is explained by a relatively high prevalence of disabling perinatal depressive symptoms [82] and wide treatment gap in the study area [83] supporting the importance of contextualising psychological interventions. The feasibility of trial procedures such as setting eligibility criteria, randomisation, masking and acceptability of outcome assessments is explained by care taken to plan a study design acceptable to this context [46].

Challenges that might have affected the feasibility of the intervention and its preliminary effectiveness include the following: (1) women's attribution of symptoms to psychosocial stressors, (2) women becoming distracted by children present during sessions, (3) women's confidence to discuss emotions and problems in Amharic and/or while being recorded, (4) HCWs' focus on recording women's worries instead of listening, and (5) the cultural practice of postnatal confinement limiting some participants from completing all four sessions or attending an outcome assessment.

PST sessions lasted longer than intended [84]. Session content may need to be reduced to enable this intervention to be integrated into ANC. Techniques such as 'worry time' and relaxation exercises were appreciated, supporting the appropriateness of the PST model. Most women attended all four intervention sessions. Few participants dropped out of the study for reasons other than moving away from the site. Feasibility of training and provider competency were evidenced by HCWs' completion and appreciation of training. Several HCWs could implement PST as planned. The intervention was safe, with no SAEs or adverse events during the trial.

However, lower than expected competency assessments support previous findings in the same setting in Ethiopia [85]. This finding suggests the need to implement inclusion criteria or post-training assessments for HCWs, or develop a more extensive apprenticeship training model. Ensuring provider fidelity and competency remains a common challenge of task-sharing. Our finding of poorer than expected competency, despite accelerated case training and specialist supervision, supports previous studies in Ethiopia [85]. However, our findings about improved provider skills across sessions are supported by similar findings in Kenya [86] demonstrating the role of experience and practice-based training in the development of new skills.

Internalising the structure and concepts of PST may require more intensive training and supervision. However, COVID-19-related pressures on ANC HCWs in the area were barriers. Besides, faster than anticipated recruitment of eligible participants led to referral of participants for sessions as soon as HCWs had completed accelerated cases, limiting the time available for in-depth supervision, reflection, and learning from those cases. Long trainer demonstrations of role plays could be improved by introducing video recorded role play demonstrations followed by more time for trainee role play practice.

The follow up rate might have been improved by narrower gestational eligibility criteria to avoid later sessions coinciding with labour and post-partum confinement. Although this feasibility study was not powered to evaluate effectiveness, improvements in mean preliminary clinical outcome scores such as GAD-7 and PCL-5 and hypothesised mediators favoured the PST arm over EUC. Positive preliminary effectiveness of the intervention may have arisen from improved HCW skills in empathetic, supportive, and compassionate care. Despite a small sample, bigger improvements in the intervention arm

in anxiety and PTSD symptoms relative to the observed reduction in depressive symptoms is an interesting finding indicating the potential for improved problem-solving skills to have enhanced women's self-efficacy, improving anxiety symptoms.

Potential areas to improve the feasibility of PST and its preliminary effectiveness are (1) planning to address low income and engage husbands so as to reduce the extent to which problems of poverty and marital conflict remain unresolved. For example, husbands could care for children while women receive PST. (2) Adapting the intervention for delivery in other local languages and potentially omitting audio recordings to enhance disclosure and acceptability. (3) Narrowing eligibility criteria to 12–30 weeks to allow the course of sessions to be completed before the due date. (4) Reducing session content to shorten session duration and enhance the feasibility of ANC integration. (5) Consider selection, post-training assessment or apprenticeship models to enhance therapist competency and (6) enhance access to training materials through video recordings of role play demonstrations.

This study primarily aimed to evaluate feasibility for a future trial and was not powered to assess efficacy. Our results are influenced by cultural and language aspects of our rural setting, where Amharic is spoken. Exploration is required in urban areas and other cultural contexts of Ethiopia. The requirement to compensate HCWs financially was essential in this low-resource setting but may have disproportionately incentivised participation.

## Conclusion

Despite the low-resource setting and very limited prior training, four sessions of problem-solving therapy were feasible and acceptable to deliver in ANC in rural Ethiopia. A randomised, controlled trial design was feasible and acceptable to implement, raising the potential for a fully powered trial to determine the efficacy of appropriately trained, supervised HCW-delivered PST for antenatal depressive symptoms in rural Ethiopia.

### Abbreviations

| | |
|---|---|
| ANC | Antenatal care |
| CDT-Africa | Center for Innovative Drug Development for Therapeutic Trials–Africa |
| DSM-IV | Diagnostic Statistical Manual–IV |
| EUC | Enhanced usual care |
| ENACT | Enhancing assessment of common therapeutic factors |
| GAD-7 | Generalized disorder anxiety |
| HEWs | Health extension workers |
| MINI | Mini International Neuropsychiatric Interview |
| MhGAP | Mental Health Gap Action Programme |
| PCL-5 | PTSD Checklist for DSM-5 |
| PHC | Primary health care |
| PHQ-9 | Patient health questionnaire-9 |
| PST | Problem solving therapy |
| PRIME | Programme for Improving Mental Health Care |
| ODK | Open Data Kit (programme) |
| OSS-3 | Oslo Social Support with 3 items |
| RCT | Randomised controlled trial |
| SNNPR | Southern nations and nationalities and peoples' region |
| SAE | Serious adverse event |
| WHODAS | World health organization disability scale |

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### Authors' contributions

Conceived the study (TB, RK, HC, KS, BM, FG); revised the protocol (TB, RK, HC, KS, BM, SH, GM, LH), took part in data collection: TB, RK, HC, KS, BM, EF, AM; took part in data analysis TB, RK, HC, KS, BM, GM; wrote the first draft TB, RK, HC, KS, BM. All co-authors read the manuscript.

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### Data availability

The datasets generated and/or analysed during the current study are not publicly available until we could publish the forthcoming study, but are available from the corresponding author on reasonable request.

## Declarations

### Ethics approval and consent to participate

Our feasibility trial protocol was registered on the Pan-African clinical trials registry, (PACTR): registration number: PACTR202008712234907 on 18/08/2020; https://pactr.samrc.ac.za/TrialDisplay.aspx?TrialID=9578. We received ethical approval from the scientific committee of CDT-Africa and the Institutional Review Board of Addis Ababa University, College of Health Sciences (IRB reference: 049/19/CDT). All participants provided written (thumbprint for non-literates) informed consent.

### Consent for publication

Not applicable.

### Competing interests

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**References**

1. WHO. Depression and other common mental disorders: global health estimates. Geneva: WHO. 2017.
2. Chowdhary N, Sikander S, Atif N, Singh S, Fuhr D, Rahman A, et al. The content and delivery of psychological interventions for perinatal depression by non-specialist health workers in low and middle income countries: a systematic review. Best Pract Res Clin Obstet Gynaecol. 2014;28:113–33.
3. Fisher J, Mello M, Patel V, Rahman A, Tran T, Holtn S, et al. Prevalence and determinants of common perinatal mentala disorders in low income and lower middle income countries: a systematic review. Bull World Health Organ. 2012;90:139–49.
4. Peltzer K, Szrek H, Ramlagan S, Leite R, Chao L. Depression and social functioning among HIV-infected and uninfected persons in South Africa. AIDS Care. 2015;27(1):41–6.
5. Senturk V, Hanlon C, Medihin G, Dewey M, Araya M, Alem A, Prince M, Stewart R. Impact of perinatal somatic and common mental disorder symptoms on functioning on Ethiopian women: the P-MaMiE population based cohort study. J Affect Disord. 2012;136:340–9.
6. Bindt C, Appiah-Poku J, Te Bonle M, Schoppen S, Feldt T, Barkmann C, et al. Antepartum depression and anxiety associated with disability in African women: cross-sectional results from the CDS study in Ghana and Cote d'Ivoire. PLoS ONE. 2012;7(10):48396.
7. Katon J, Russo J, Gavin A, Melville J, Katon W. Diabetes and depression in pregnancy: is there an association? J Women's Health. 2011;20(7):983.
8. Katon W. Epidemiology and treatment of depression in patients with chronic medical illness. Dialogues Clin Neurosci. 2011;13(1):7–23.
9. Katon W, Russo J, Melville J, Katon J, Gavin A. Depression in pregnancy is associated with pre-existing but not pregnancy-induced hypertension. Gen Hosp Psychiatry. 2012;34(1):9–16.
10. Heyningen T, Myer L, Onah M, Tomlinson M, Field S, Honikman S. Antenatal depression and adversity in urban South Africa. J Affect Disord. 2016;203:121–9.
11. Hanlon CMG, Alem A, Araya M, Abdulahi A, Tesfaye M, Wondimagegn D, Patel V, Prince M. Detecting perinatal common mental disorders in Ethiopia: validation of self reporting questionaire and Edinburgh postnatal depression scale. J Affect Disord. 2008;108:251–62.
12. Tesfaye M, Hanlon C, Wondimagegn D, Alem A. Detecting posnatal common mental disorders in Addis Ababa, Ethiopia: validation of the Edinburgh postnatal depression scale and Kessler scales. J Affect Disord. 2009;122:102–8.
13. Medihin G, Hanlon C, Dewey M, Alem A, Tesfaye F, Lakew Z, et al. The effect of maternal common mental disorders on infant undernutrition in Butajira, Ethiopa: P-MaMiE study. BMC Psychiatry. 2010;10:32.
14. Saeed A, Raana T, Saeed AM, Humayun A. Effect of antenatal depression on maternal dietary intake and neonatal outcome: a prospective cohort. Nutr J. 2016;15(1):64.
15. Bitew T, Hanlon C, Kebede E, Honikman S, Fekadu A. Antenatal depressive symptoms and perinatal complication: a prospective study in rural Ethiopia. BMC Psychiatry. 2017;17(301):1.
16. Bitew T, Hanlon C, Kebede E, Medihn G, Fekadu A. Antenatal depressive symptoms and maternal health care utilisation: a population-based study of pregnant women in Ethiopia. BMC Pregnancy Childbirth. 2016;16:301.
17. Bitew T, Hanlon C, Kebede E, Honikman S, Fekadu A. Effect of ANDs on delivery and postnatal care utilisation: propective studt. BMC Pregnancy Childb. 2017;17:1186.
18. Baron E, Hanlon C, Mall S, Honikman S, Breuer E, Kathree T, et al. Mental health in primary care in five low and middle income countries: a situational analysis. BMC Health Serv Res. 2016;16(53):1293–316.
19. Sharma V, Sommerdyk C. Are antidepressants effective in the treatment of postpartum depression? A systematic review. Prim Care comparison CNC Diord. 2013;15(6):26642.
20. Kennedy SH, Rizvi S. Comparative efficacy of newer antidepressants for major depression: a Canadian perspective. Can J Diagn. 2009.
21. Udechuku A, Nguyen T, Hill R, Szego K. Antidepressants in pregnancy: a systematic review. Aust N Z J Psychiatry. 2010;44(11):978–96 PubMed PMID: 21034181. Epub 2010/11/03. eng.
22. Gadot Y. Koren G. The use of antidepressants in pregnancy: focus on maternal risks J Obstet Gynaecol Can. 2015;37(1):56–63.
23. Solomon A, Haaga DA, Arnow BA. Is clinical depression distinct from subthreshold depressive symptoms? A review of the continuity issue in depression research. J Nerv Ment Dis. 2001;189(8):498–506 PubMed PMID: 11531201. Epub 2001/09/04. eng.
24. Klein DN. Classification of depressive disorders in the DSM-V: proposal for a two-dimension system. J Abnorm Psychol. 2008;117(3):552–60 PubMed PMID: 18729608. eng. 2008/08//.
25. Patel V, Chowdhary N, Rahman A, Verdeli H. Improving access to psychological treatments: lessons from developing countries. Behaviour research and therapy. 2011:1–6.
26. Munodawafa M, Mall S, Lund L, Schneider M. Process evaluations of task sharing interventions for perinatal depression in low and middle income countries (LMIC): a systematic review and qualitative meta-synthesis. BMC Health Serv Res. 2018;18:1.
27. Rahman A, Fisher F, Bower P, Luchters S, Tran T, Yasamy T, et al. Interventions for common perinatal mental disorders in women in low- and middle-income countries: a systematic review and meta-analysis Bull World Health Organ. 2013;91:593–601.
28. Singla D, Kohrt B, Murray L, Anand A, Chorpita B, Patel V. Psychological treatments for the world: lessons from low- and middle-income countries: annual review of clinical psychology. Annu Rev Clin Psychol. 2017;13:149–81.
29. Azale T, Fekadu A, Hanlon C. Treatment gap and help-seeking for postpartum depression in a rural African setting. BMC Psychiatry. 2016;16(196):1.
30. FMOH. National mental health strategy of Ethiopia. Addis Ababa. 2012. 2012/13 -2015/16.
31. Chowdhary N, Jotheeswaran A, Nadkarni A, Hollon S, King M, Jordans M, et al. The methods and outcomes of cultural adaptations of psychological treatments for depressive disorders: a systematic review. Psychol Med. 2014;44:1131–46.
32. Bitew T, Keynejad R, Honikman H, Sorsdahl K, Myers B, Fekadu A, et al. Stakeholder perspectives on antenatal depression and the potential for psychological intervention in rural Ethiopia: a qualitative study. BMC Pregnancy Childb. 2020;20(371):1.
33. Azale T, Fekadu A, Hanlon C. Postpartum depressive symptoms in the context of high social adversity and reproductive health threats: a population-based study. Int J Ment Health Syst. 2018;12(1):42 2018/07/28.
34. Bitew T, Keynejad R, Myers B, Honikman S, Sorsdahl K, Hanlon C. Adapting an intervention of brief problem-solving therapy to improve the health of women with antenatal depressive symptoms in primary healthcare in rural Ethiopia. PFS. 2022;8:202.
35. Sorsdahl K, van der Westhuizen C, Cuijpers P, Pool M, & JY, Myers B. PROJECT MIND: mental well-being programme for adults experiencing symptoms of depression, anxiety and alcohol use problems (generic field-trial version 1.1). South Africa: Alan J. Flisher Centre for Public Mental Health. 2023.
36. Sorsdahl K, Stein D, Carrara H, Myers B. Problem solving styles among peoplewho use alcohol and other drugs in South Africa. Addict Behav. 2014;39:122–6.

37. Bell A, D'Zurilla T. Problem-solving therapy for depression: a meta-analysis☆. Clin Psychol Rev. 2009;29:348–53.
38. Malouff J, Thorsteinsson E, Schutte N. The efficacy of problem solving therapy in reducing mental and physical health problems: a meta-analysis. Clin Psychol Rev. 2007;27:46–57.
39. Myers B, Lombard CJ, Lund C, Joska JA, Levitt N, Naledi T, Petersen Williams P, van der Westhuizen C, Cuijpers P, Stein DJ, Sorsdahl KR. Comparing dedicated and designated approaches to integrating task-shared psychological interventions into chronic disease care in South Africa: a three-arm, cluster randomised, multicentre, open-label trial. Lancet (London, England). 2022;400(10360):1321–33.
40. Chibanda D, Mesu P, Kajawu L, Cowan F, Araya R, Abas M. Problem-solving therapy for depression and common mental disorders in Zimbabwe: piloting a task-shifting primary mental health care intervention in a population with a high prevalence of people living with HIV. BMC Public Health. 2011;11:828.
41. Areán P, Raue P, McCulloch C, Kanellopoulos D, Seirup J, Banerjee S, et al. Effects of problem-solving therapy and clinical case management on disability in low-income older adults. Am J Geriatr Psychiatry. 2015;23:1307e14.
42. Hof E, Stein D, Marks I, Tomlinson M, Cuijpers P. The effectiveness of problem solving therapy in deprived South African communities: results from a pilot study. BMC Psychiatry. 2011;11:156.
43. Craig P, Dieppe P, Macintyre S, Michie S, Nazareth I, Petticrew M. Developing and evaluating complex interventions: new guidance: new guideline. In: Council MR, editor. UK: Medical Research Council. 2008.
44. Wingood G, DiClemente R. The ADAPT-ITT model: a novel method of adapting evidence-based HIV interventions. J Acquir Immune Defic Syndr. 2008;47(supplement 1):S40.
45. Lund CTM, De Silva M, Fekadu A, Shidhaye R, Jordans M, et al. PRIME: a programme to reduce the treatment gap for mental disorders in five low- and middle-income countries. PLoS Med. 2012;9(12):e1001359.
46. Bitew T, Keynejad R, Myers B, Honikman S, Medhin G, Girma F, et al. Brief problem-solving therapy for antenatal depressive symptoms in primary care in rural Ethiopia: protocol for a randomised, controlled feasibility trial. Pilot Feasibility Stud. 2021;7:35.
47. Keynejad R, Bitew T, Sorsdahl K, Myers B, Honikman S, Medhin G, et al. Problem solving therapy (PST) tailored for intimate partner violence (IPV) versus standard PST and enhanced usual care for pregnant women experiencing IPV in rural Ethiopia: protocol for a randomised controlled feasibility trial. Trials. 2020;21:454.
48. Keynejad RC, Bitew T, Sorsdahl K, Myers B, Honikman S, Medhin G, Deyessa N, Sevdalis N, Tol WA, Howard L, Hanlon C. Problem-solving therapy for pregnant women experiencing depressive symptoms and intimate partner violence: a randomised, controlled feasibility trial in rural Ethiopia. PLOS Glob Public Health. 2023;3(10):454.
49. Hanlon C, Medhin GC, Selamu M, Breuer E, Worku B, Hailemariam H, et al. Validity of brief screening questionnaires to detect depression in primary care in Ethiopia. J Affect Disord. 2015;186:32–9.
50. Roaldset J, Linaker O, Bjørkly S. Predictive validity of the MINI suicidal scale for self-harm in acute psychiatry: a prospective study of the first year after discharge. Arch Suicide Res. 2012;16:287–302.
51. World Health Organization. mhGAP Intervention Guide Mental Health Gap Action Programme for mental, neurological and substance use disorders in non-specialized health settings. In: Abuse DoMHaS, editor.: World Health Organization; 2016.
52. Fekadu A, Hanlon C, Medhin G, Alem A, Selamu M, Giorgis TW, et al. Development of a scalable mental healthcare plan for a rural district in Ethiopia. Br J Psychiatry. 2016;208:s4.
53. Abbas Tavallaii S, Ebrahimnia M, Shamspour N, Assari S. Effect of depression on health care utilization in patients with end-stage renal disease treated with hemodialysis. Eur J Intern Med. 2009;20(4):411–4 PubMed PMID: 19524185. Epub 2009/06/16. eng.
54. Viechtbauer W, Smits l, Kotz D, Budé L, Spigt M, Serroyen J, et al. A simple formula for the calculation of sample size in pilot studies. J Clin Epidemiol. 2015;68(11):1375–9.
55. Zhang A, Park S, Sullivan JE, Jing S. The effectiveness of problem-solving therapy for primary care patients' depressive and/or anxiety disorders: a systematic review and meta-analysis. J Am Board Fam Med : JABFM. 2018;31(1):139–50 PubMed PMID: 29330248. Epub 2018/01/14. eng.
56. Kirkham JG, Choi N, Seitz DP. Meta-analysis of problem solving therapy for the treatment of major depressive disorder in older adults. Int J Geriatr Psychiatry. 2016;31(5):526–35 PubMed PMID: 26437368. Epub 2015/10/06. eng.
57. Hegel M, Arean P. Problem solving treatment for primary care (PST-PC): A treatment manual for depression. 2003.
58. Kohrt B, Jordans M, Rai S, Shrestha P, Luitel N, Ramaiuya M, et al. Therapist competence in global mental health: development of the enhancing assessment of common therapeutic factors (ENACT) scale. Bhavr Res Ter. 2015;69:11–21.
59. Kohrt B, Ramaiya M, Bhardwaj A, Jordans M. Development of a scoring systeme for non-specialits rating of clincial competernce in global mental health: a qualtiative process evaluation of the enhancing assessment of common therapeutic factors (ENACT) scale. Global Mental Health. 2015;2(e23):1–6.
60. JBA. Evaluation brief: measuring implementation fidelity. In: Associates JB, editor. Arlington, VA: Author. 2009.
61. Feyissa Y, Hanlon C, Emyu S, Cornick R, Fairall L, Gebremichael D, et al. Using a mentorship model to localise the Practical Approach to Care Kit (PACK): from South Africa to Ethiopia. BMJ Glob Health. 2019;3:e001108.
62. FMoH. Ethiopia primary healthc are clinical juidelines: care of children 5–14 years and adulte 15 years or older in health centeres. In: Ethiopia Fdro, editor. Addis Ababa: Federal democratice republic of Ethiopia. 2017.
63. Gureje O, Oladeji BD, Araya R, et al. Expanding care for perinatal women with depression (EXPONATE): study protocol for a randomized controlled trial of an intervention package for perinatal depression in primary care. BMC Psychiatry. 2015;15:136.
64. Gureje O, Oladeji B, Montgomery A, Araya R, Bello T, Chisholm D, et al. High- versus low-intensity interventions for perinatal depression delivered by non-specialist primary maternal care providers in Nigeria: cluster randomised controlled trial (the EXPONATE trial). Brit J Psychiatry. 2019;215(3):528-35. https://doi.org/10.1192/bjp.2019.4.2019.
65. Schulz KF AD, Moher D; CONSORT Group. CONSORT 2010 Statement: updated guidelines for reporting parallel group randomised trials. BMC Med. 2010;24(8):18.
66. Kroenke K, Spitzer R, Wiliams J. Validity of a brief depression severity measure. J Gen Intern Med. 2001;16:606–13.
67. Dejusu R, Vickers K, Melin G, Williams M. A system-based approach to depression management in primary care using the patient health questionnaire-9. Mayo Clin Proc. 2007;82(11):1395–402.
68. McIntyre R, Fallu A, Konarski J. Measurable outcomes in psychiatric disorders: remission as a marker of wellness. Clin Ther. 2006;28(11):1882.
69. Girma F, Detecting depression during pregnancy: validation of PHQ-9, Kessler-10, Kessler-6 and SRQ-20 in Butajira area health centers antenatal care clinics, Ethiopia. Addis Ababa University. Ethiopia. Addis Ababa: Addis Ababa University; 2013. p. 2013.
70. Spitzer R, Kroenke K, Williams J, et al. A brief measure for assessing generalized anxiety disorder the GAD-7. Arch Intern Med. 2006;166(10):1092–7.
71. WHO. Measuring health and disability manual for WHO disability assessment schedule: WHODAS 2.02010.
72. Yohannes K, Gebeyehu A, Adera T, Ayano G, Fekadu W. Prevalence and correlates of post-traumatic stress disorder among survivors of road traffic accidents in Ethiopia. Int J Ment Heal Syst. 2018;12(1):1–8.
73. Blevins C, Weathers F, Davis M, Witte T, Domino J. The posttraumatic stress disorder checklist for DSM-5 (PCL-5): development and initial psychometric evaluation. J Trauma Stress. 2015;28(6):489–98.
74. Zink T, Levin L, Putnam F, Beckstrom A. Accuracy of five domestic violence screening questions with nongraphic language. Clin Pediatr. 2007;46(2):127–34.
75. C. GM. WHO multi-country study on women's health and domestic violence against women: initial results on prevalence, health outcomes and women's responses. 2005. http://www.who.int/reproductivehealth/publications/violence/24159358X/en/.
76. García-Moreno C. WHO multi-country study on women's health and domestic violence against women: initial results on prevalence, health outcomes and women's responses 2005. 2005. Available from: http://www.who.int/reproductivehealth/publications/violence/24159358X/en/.
77. Fok CCTAJ, Henry D, Mohatt GV. Multicultural Mastery Scale for Youth: multidimensional assessment of culturally mediated coping strategies. Psychol Assess. 2012;24(2):313.

78. Brugha T, Cragg D. The list of threatening experiences: the reliability and validity of a brief life events questionnaire. Acta Psychiatr Scand. 1990;82(1):77–81.
79. Sorato B. Client service reciept inventory. 2003.
80. Berg L, Beutel M, Hinz A, Zenger M, Härter M, Nater U, Brähler E. Social support in the general population: standardization of the Oslo social support scale (OSSS-3). BMC Psychol. 2018;6(1):1–8.
81. Craig P, Di Ruggiero E, Frohlich KL, Mykhalovskiy E and White M, on behalf of the Canadian Institutes of Health Research (CIHR)–National Institute for Health Research (NIHR) Context Guidance Authors Group. Taking account of context in population health intervention research: guidance for producers, users and funders of research. Southampton: NIHR Evaluation, Trials and Studies Coordinating Centre; 2018.
82. Zegeye A, Alebe A, Gebrie A, Tesfaye B, Abebe Y, Adane F, et al. Prevalence and determinants of antenatal depression among pregnant women in Ethiopia: a systematic review and metaanalysis. BMC Pregnancy Childb. 2018;18:462.
83. Azale T, Fekadu A, Hanlon C. Treatment gap and help-seeking for postpartum depression in a rural African setting. BMC Psychiatry. 2016;16:196.
84. Bitew T, Keynejad R, Myers B, Honikman S, Sorsdahl K, Hanlon C. Adapting an intervention of brief problem-solving therapy to improve the health of women with antenatal depressive symptoms in primary healthcare in rural Ethiopia. Pilot Feasibility Stud. 2022;8(1):202 2022/09/09.
85. Asher L, Birhane R, Teferra S, Milkias B, Worku B, Habtamu A, et al. "Like a doctor, like a brother": achieving competence amongst lay health workers delivering community-based rehabilitation for people with schizophrenia in Ethiopia. PLoS ONE. 2021;16(2):e0246158.
86. Giusto A, Vander Missen MR, Kosgei G, Njiriri F, Puffer E, Kamaru Kwobah E, et al. Peer-delivered problem-solving therapy for adolescent mental health in Kenya: adaptation for context and training of peer-counselors. Res Child Adolescent Psychopathol. 2023;51(9):1243–56 2023/09/01.

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